

Control work 3, and work on the mistakes

Perpendicularity in space in one paper — where naming the two intersecting lines is worth as much as the answer.

LESSON 43–44

2 × 40 MIN

GEOMETRY 10, NAZORAT ISHI 3

IGCSE REVIEW

LESSON CLOCK

3

40'

12'

23'

12'

Instructions	3 min
The paper	40 min
Answers	12 min
Rewrite	23 min
The test drill	12 min

LEARNING OBJECTIVES

- Apply the perpendicularity theorems under time.
- Quote a test by name and verify its conditions.
- Classify each lost mark and rewrite the solution.
- Consolidate the three perpendiculars theorem.

TEXTBOOK

National	pp. 207–210
Cambridge	Core & Extended, pp. 345–346
Workbook	Control paper G3

01

Explanation

The paper — 25 marks, 40 minutes

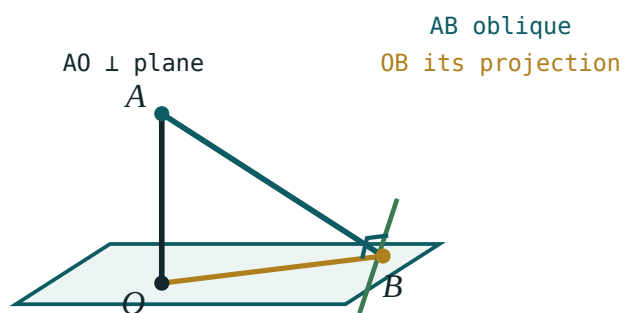
Q	TASK	MARKS	FROM
1	Prove $DD_1 \perp AC$ in a cube, quoting the test	4	L33–34
2	A point is 9 above a plane; two obliques have projections 12 and 40. Find both	4	L35–37
3	An oblique of 20 makes 30° with the plane. Find the perpendicular and the projection	4	L35–37
4	Prove $AC_1 \perp BD$ in a cube by the three perpendiculars	5	L38–39
5	A regular pyramid, base 10, height 12: find the slant height and the face-to-base dihedral angle	5	L38–42
6	Cube of edge 6: find the distance from A to the plane BDD_1B_1	3	L35–37

THREE MARKS FOR NAMING

Q1 gives one mark for naming the two intersecting lines, Q4 one for identifying the projection, Q5 one for saying why the arm is perpendicular to the edge. Three of twenty-five.

The three errors

ERROR	LOOKS LIKE	KIND
parallel lines used in the test	" $AA_1 \perp AB$ and $AA_1 \perp D_1C_1$, so $AA_1 \perp ABCD$ "	knowledge
wrong projection	the projection of AC_1 given as AB	method
arm not $\perp$ to the edge	a dihedral angle measured between any two rays	method
slant height and lateral edge confused	OM used where OA was needed	careless



a line of the plane through B is $\perp$ AB
exactly when it is $\perp$ OB

Questions 4 and 5 in one picture.

The test drill

Eight statements on the board. For each, the class names the **test or theorem** that settles it, in three seconds:

STATEMENT	TOOL
$AA_1 \perp ABCD$	test for line $\perp$ plane
$AC_1 \perp BD$	three perpendiculars
$ABB_1A_1 \perp ABCD$	test for perpendicular planes
$SM \perp AB$ in a pyramid	three perpendiculars
the shortest segment to a plane	the perpendicular

02 Worked examples

EXAMPLE 1 Model answer, Q4: prove $AC_1 \perp BD$ in a cube.

- 1 $CC_1 \perp ABCD$
So AC is the projection of AC_1 .
- 2 $BD \perp AC$
Diagonals of a square.
- 3 $BD \subset ABCD$ through the foot C .
- 4 Three perpendiculars.

ANSWER

$$AC_1 \perp BD$$

EXAMPLE 2 Model answer, Q5: pyramid base 10, height 12.

- 1 $OM = 5$
Half the base edge.
- 2 $SM = \sqrt{144 + 25} = 13$
The slant height.
- 3 $OM \perp AB$, and $SM \perp AB$ by the theorem.
The linear angle is $\angle SMO$.
- 4 $\tan \theta = \frac{12}{5} \Rightarrow \theta \approx 67.4^\circ$

ANSWER

$$SM = 13; \theta \approx 67.4^\circ$$

EXAMPLE 3 A learner used $AA_1 \perp AB$ and $AA_1 \perp D_1C_1$ to conclude $AA_1 \perp ABCD$. Name the mistake.

- ① D_1C_1 does not lie in $ABCD$.
- ② And it is parallel to AB , not intersecting.
Two failures at once.
- ③ Correct: use AB and AD .

ANSWER

The two lines must lie in the plane and intersect

03 Interactive model

Run the test drill before the rewrite, holding a real box.

Name the tool

To prove $AA_1 \perp ABCD$ use:

three perpendiculars

the test for line $\perp$ plane

the definition

Pythagoras

To prove $AC_1 \perp BD$ use:

three perpendiculars

the test for planes

the definition

measurement

Perpendicular 9, projection 12: the oblique is:

15

21

3

108

Oblique 20 at 30° : the perpendicular is:

Pyramid base 10, height 12: the slant height is:

Cube edge 6: distance from A to BDD_1B_1 :

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Test (criterion)	Alomat	Признак
Three perpendiculars	Uch perpendikulyar	Три перпендикуляра
Dihedral angle	Ikki yoqli burchak	Двугранный угол
Linear angle	Chiziqli burchak	Линейный угол
Justification	Asoslash	Обоснование
Careless error	E'tiborsizlik xatosi	Ошибка по невнимательности
Correction	Tuzatish	Исправление

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The test for a line $\perp$ a plane needs:

one line

two intersecting lines of the plane

two parallel lines

three points

The three perpendiculars theorem relates:

two planes

an oblique and its projection

two obliques

two normals

A dihedral angle's arms must be $\perp$ to:

each other

the edge

the base

the normal

The shortest segment from a point to a plane is:

any oblique

the perpendicular

the projection

the diagonal

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Prove $DD_1 \perp AC$

ANSWER $DD_1 \perp ABCD$ and $AC \subset ABCD$

2. Perpendicular 9, projection 12: the oblique

ANSWER 15

3. Perpendicular 9, projection 40: the oblique

ANSWER 41

4. Oblique 20 at 30° : the perpendicular

ANSWER 10

5. Same: the projection

ANSWER $10\sqrt{3}$

6. Pyramid base 10, height 12: slant height

ANSWER 13

7. Cube edge 6: distance from A to BDD_1B_1

ANSWER $3\sqrt{2}$

Medium · the standard question

7 PROBLEMS

1. Prove $AC_1 \perp BD$ by the three perpendiculars

ANSWER Projection AC , and $BD \perp AC$

2. Pyramid base 10, height 12: face-to-base angle

ANSWER $\approx 67.4^\circ$

3. Pyramid base 10, height 12: lateral edge

ANSWER $\sqrt{194} \approx 13.9$

4. Cube edge 6: distance from A to A_1BD

ANSWER $2\sqrt{3}$

5. Oblique 25, projection 24: the perpendicular

ANSWER 7

6. Prove $BCC_1B_1 \perp ABCD$

ANSWER $BB_1 \perp ABCD$ lies in it

7. Cube: angle of AC_1 with the base

ANSWER $\approx 35.26^\circ$

Hard · stretch and reason

7 PROBLEMS

1. Cube edge a : distance from A to the plane A_1BD

ANSWER $\frac{a\sqrt{3}}{3}$

2. Pyramid base 12, lateral edge 10: height and both dihedral angles

ANSWER $\sqrt{82}$, $\approx 56.4^\circ$ and $\approx 46.9^\circ$

3. Regular tetrahedron edge a : the dihedral angle between faces

ANSWER $\approx 70.5^\circ$

4. Cube edge 8: distance from B to the line A_1D_1

ANSWER $8\sqrt{2}$

5. A point 15 above a plane, two obliques at 30° and 45° : their lengths

ANSWER 30 and $15\sqrt{2}$

6. Prove the linear angle is independent of the point chosen on the edge

ANSWER Respectively parallel arms

7. Cube: prove $AC_1 \perp A_1BD$

ANSWER Two of its lines are $\perp$ to AC_1

07 Homework

Homework — 4 tasks

Task 1 is the rewrite. Name the test on every proof.

1. Rewrite in full every question that lost a mark, naming the test used.
2. Five problems from the section your knowledge column was heaviest in.
3. A regular square pyramid has base edge 16 and height 15. Find the slant height, the lateral edge and both dihedral angles.

4. Cube of edge 10 : find the distance from A to the plane A_1BD .